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Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE/ NASIONALE SENIOR SERTIFIKAAT

GRADE/GRAAD 12

**MATHEMATICAL LITERACY P1/
WISKUNDIGE GELETTERDHEID V1**

NOVEMBER 2022

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Symbol/Kode	Explanation/Verduideliking
M	Method/Metode
MA	Method with accuracy/Metode met akkuraatheid
MCA	Method with consistent accuracy/Metode met volgehoue akkuraatheid
CA	Consistent accuracy/Volgehoue akkuraatheid
A	Accuracy/Akkuraatheid
C	Conversion/Herleiding
S	Simplification/Vereenvoudiging
RT	Reading from a table/graph/document/diagram/Lees vanaf tabel/grafiek/dokument/diagram
SF	Correct substitution in a formula/Korrekte vervanging in 'n formule
O	Opinion/Explanation/Opinie/Verduideliking
P	Penalty, e.g. for no units, incorrect rounding off, etc./Penalisasie, bv. vir geen eenhede, verkeerde afronding, ens.
NPR	No penalty for correct rounding/Geen penalisasie vir korrekte afronding nie
NPU	No penalty for omitting unit, but wrong unit is penalised/Geen penalisasie indien die eenheid uitgelos is, maar wel indien 'n verkeerde eenheid gebruik word.
AO	Answer only/Slegs antwoord

**These marking guidelines consist of 20 pages and 2 pages of notes.
Hierdie nasienriglyne bestaan uit 20 bladsye en 2 bladsye met notas.**

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however it stops at the second calculation error.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- Rounding is an independent mark.
- General principle of marking, if the candidate makes one mistake he loses one mark.
- A conclusion mark can only be given if relevant calculations precedes it.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas; dit hou egter op by die tweede berekeningsfout.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek, tabel, uitlegplan en kaart geneem en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- Afronding tel as 'n afsonderlike punt.
- Die algemene beginsel van merk as 'n leerder een fout maak verloor hy een punt.
- 'n Gevolgtrekkingspunt kan slegs gegee word indien relevante berekeninge dit voorgaan.

QUESTION/VRAAG 1 [30 MARKS/PUNTE] ANSWER ONLY FULL MARKS			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
1.1.1	Numerical / Numeriese ✓✓A	2A correct classification (2)	D L1 E
1.1.2	✓RT R11,99; R18,99; R39,99; R44,99; R54,99; R159,99 ; R169,99 ✓A	1RT all correct values 1A ascending order (2)	D L1 E
1.1.3	B ✓✓RT	2RT correct store (2)	D L1 E
1.1.4	White socks/Wit kouse = $\frac{R85,99}{5}$ ✓MA = R17,198 ✓A = R17,20 ✓R	1MA dividing by 5 1A price per pair 1R 2 decimal places (3)	F L1 E
1.1.5	Total cost / Totale koste ✓RT P = R110,00 + R163,00 + R186,00 + R40,50 + R85,00 + R349,00 + R318,00 = R1 251,50 ✓A	1RT correct values 1A simplification (2)	F L1 E
1.1.6 (a)	✓A The chance/likelihood of selecting Store C. ✓A Die kans/waarskynlikheid om Winkel C te kies.	1A chance/likelihood 1A store C (2)	P L1 E

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
1.1.6 (b)	$0,3333333333 \times 100\%$ ✓MA $= 33,33333333\%$ $= 33\%$ ✓A	1MA calculating percentage 1A rounded percentage (2)	P L1 E
1.2.1	✓A An investment is any form of saving that you put into a financial scheme, bank or stokvel that will result in interest. ✓A <i>'n Belegging is 'n vorm van spaar waar jy geld in 'n finansiële skema, bank of stokvel sit wat sal lei na rente.</i>	1A form of savings 1A interest (2)	F L1 E
1.2.2	Total contributions/Totale bydrae $R2\ 500 \times 24$ ✓MA $= R60\ 000$ ✓A	1MA multiply by months 1A simplification (2)	F L1 E
1.2.3	Interest earned/Rente verdien ✓MA $R92\ 065,71 - R60\ 000$ $= R32\ 065,71$ ✓A	1MA subtract correct values 1A simplification (2)	F L1 E
1.2.4	More interest/Meer rente ✓MA $R92\ 065,71 - R74\ 286,84$ $= R17\ 778,87$ ✓A	1MA subtracting correct values 1A simplification (2)	F L1 E
1.3.1	✓A Compound / Triple / Grouped / Multiple / Clustered bar graph ✓A <i>Saamgestelde/ Drievoudige/ Gegroepeerde / Veelvoudige staafgrafiek</i>	1A type 1A bar graph (2)	D L1 E
1.3.2	95 ULP / Unleaded petrol / ULP / 95 95 ULP / Loodvrye petrol / ULP / 95 ✓✓A	2A correct product (2)	D L1 M
1.3.3	In rand /In rand $1\ 955,28 \text{ c/ℓ} \div 100$ ✓MA $= R19,55/\text{ℓ}$ ✓A $= R19,50/\text{ℓ}$ ✓R	1MA dividing by 100 1A rand per litre 1R rounding (3)	F L1 M
		[30]	

QUESTION/VRAAG 2 [34 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.1.1	Policy number / <i>Polisnommer</i> = 23388350 ✓✓RT	2RT correct number (2)	F L1 E
2.1.2	Table values excluding A and discount / <i>Tabel waardes uitsluitend A en die afslag</i> $R7,16 + R200,41 + R520,41 + R133,16 + R201,79 + R23,30 + R9,07$ ✓MA = R1 095,30 Value of A / <i>Waarde van A</i> ✓MCA $R2\ 184,21 - R1\ 095,30 + R266,15$ ✓MA = R1 355,06 ✓CA OR/OF ✓MA $R1\ 095,30 + A + (-R266,15) = R2\ 184,21$ ✓MA ✓MCA $A = R2\ 184,21 + R266,15 - R1\ 095,30$ = R1 355,06 ✓CA OR/OF Table values – A = / <i>Tabel waarde – A =</i> ✓MA $(R7,16 + R200,41 + R520,41 + R133,16 + R201,79 + R23,30 + R9,07) - R266,15$ ✓MA = R829,15 $A = R2\ 184,21 - R829,15$ ✓MCA = R1 355,06 ✓CA	1MA adding all premiums 1MCA subtracting from total premium 1MA adding the discount 1CA simplification OR/OF 1MA adding all premiums 1MCA subtracting from total premium 1MA adding the discount 1CA simplification OR/OF 1MA adding all premiums 1MA subtracting the discount 1MCA subtracting from total premium 1CA simplification (4)	F L3 D
2.1.3	Percentage discount / <i>Persentasie afslag</i> ✓RT $\frac{R266,15}{R2\ 450,36} \times 100\%$ ✓MA = 10,86% ✓CA	1RT correct amount 1MA correct %-calculation 1CA simplification	F L3 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.1.3	OR/OF ✓RT $\% \text{ discount} = \frac{R2\,450,36 - R2\,184,21}{R\,2\,450,36} \times 100\% \quad \checkmark \text{MA}$ $= 10,86\% \quad \checkmark \text{CA}$ OR/OF ✓RT $\text{Percentage discount} = 100\% - \left(\frac{R2\,184,21}{R2\,450,36} \times 100\% \right)$ $= 100\% - 89,14\% \quad \checkmark \text{MA}$ $= 10,86\% \quad \checkmark \text{CA}$	OR/OF 1RT correct amount 1MA correct %-calculation 1CA simplification OR/OF 1RT correct amount 1MA correct %-calculation 1CA simplification NPR (3)	
2.1.4	Claim amount / <i>Eisbedrag</i> ✓RT $R43\,520 - R7\,000$ $= R36\,520 \quad \checkmark \text{A}$	1RT identifying R7 000 1A claim amount (2)	F L1 M
2.1.5	Amount VAT / <i>BTW-bedrag</i> ✓A $\frac{15}{115} \times \frac{R2\,184,21}{1}$ $\checkmark \text{MA}$ $= R284,90 \quad \checkmark \text{A}$ OR/OF $\text{Amount before VAT} = \frac{R2\,184,21}{1,15} \quad \checkmark \text{MA}$ $= R1\,899,31 \quad \checkmark \text{A}$ $\text{VAT amount} = R2\,184,21 - R1\,899,31$ $= R284,90 \quad \checkmark \text{A}$	1A correct VAT calculation 1MA multiplying by $\frac{15}{115}$ 1A simplification OR/OF 1MA dividing by 1,15 1A VAT excluded amount 1A simplification (3)	F L2 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.1.6	The VW Polo costs more than the Toyota Corolla. / <i>Die VW Polo is duurder as die Toyota Corolla.</i> ✓✓O OR/OF The Toyota is an older model. / <i>Die Toyota is 'n ouer model.</i> (The VW is a newer model. / <i>Die VW is 'n nuuter model.</i>) ✓✓O OR/OF The retail value of the VW Polo is higher, therefore replacement value is higher. / <i>Die herverkoopwaarde van die VW Polo is hoër, daarom is die vervangingswaarde duurder.</i> ✓✓O OR/OF The VW is a high risk vehicle. / <i>Die VW is 'n hoë risiko voertuig.</i> ✓✓O OR/OF The primary driver of the VW is a younger driver who is inexperienced as a driver. <i>Die primêre bestuurder van die VW is 'n jonger bestuurder wat 'n onervare bestuurder is.</i> ✓✓O	2O correct explanation (2)	F L4 M
2.1.7	His premium will increase as his household content value will increase to more than R200 000. / <i>Sy premie sal verhoog aangesien sy huishoudelike inhoudswaarde gevolglik na meer as R200 000 sal verhoog.</i> ✓✓O	2O valid conclusion (2)	F L4 M
2.2.1	Cost of sanitation / <i>Koste van sanitasie</i> = R228,06 ✓RT = R228,10 ✓R	AO 1RT correct amount 1R correct rounding (2)	F L2 M
2.2.2	Tariff before increase / <i>Tarief voor verhoging</i> ✓RT R16,03 – R0,66 = R15,37 ✓A Cost of sanitation / <i>Koste vir sanitasie</i> 4,1 kℓ × R15,37 ✓MCA = R63,02 ✓CA	1RT reading from table 1A simplification 1MCA multiply by tariff 1CA simplification	F L3 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.2.2	OR/OF Amount after increase / <i>Bedrag na verhoging</i> $= R16,03 \times 4,1 \text{ k}\ell$ $= R65,723 \checkmark A$ Increase / <i>Verhoging</i> $\checkmark RT$ $= R0,66 \times 4,1 \text{ k}\ell$ $= R2,706$ Amount before increase / <i>Bedrag voor verhoging</i> $= R65,723 - R2,706 \checkmark MCA$ $= R63,02 \checkmark CA$	OR/OF 1A simplification 1RT reading from table 1MCA multiply by tariff 1CA simplification (4)	
2.2.3	Sanitation Bill – Cape Town: Ms Brown <i>Sanitasierekening – Kaapstad: Me Brown</i> $\checkmark MA \quad \checkmark RT$ $4,2 \text{ k}\ell \times R16,03 = R 67,33$ $3,15 \text{ k}\ell \times R22,02 = R 69,36$ $14,65 \text{ k}\ell \times R30,92 = \underline{R452,98}$ $= R589,67 \checkmark CA$ Sanitation Bill – Johannesburg: Mr Jones <i>Sanitasierekening – Johannesburg: Mnr Jones</i> VAT / BTW: $\checkmark RT$ $= R443,96 \times \frac{15}{100}$ $= R66,59$ Total amount / Totale bedrag $= R443,96 + R66,59 \checkmark MCA$ $= R510,55 \checkmark CA$ $= R443,96 \times \frac{115}{100}$ Difference in Sanitation Bill / <i>Verskil in Sanitasierekening</i> $R589,67 - R510,55 \checkmark MCA$ $= R79,12 \checkmark CA$	1MA all (3) correct kℓ 1RT 3 tariffs 1CA finding total water bill 1RT correct amount 1MCA adding values 1CA simplification 1MCA subtracting values 1CA simplification (8)	F L3 D

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
2.2.4	The fixed rate allows you to use as much as you can for the same amount. / <i>Die vaste koers laat jou toe om soveel as wat jy kan vir dieselfde bedrag te gebruik.</i> ✓✓○ OR/OF It benefits home owners with smaller properties who pay less. / <i>Dit bevoordeel eienaars van kleiner huise wat minder gaan betaal.</i> ✓✓○ OR/OF Even if the usage of water varies /differs from month to month, the cost/bill remains the same amount. / <i>Selfs as die water gebruik verskil van maand tot maand bly die koste/rekening dieselfde.</i> ✓✓○	2O correct explanation (2)	F L4 M
		[34]	

QUESTION/VRAAG 3 [24 MARKS/PUNTE]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
3.1.1	Total value / <i>Totale waarde</i> $\checkmark$RT $= 18,4 + 5,6 + 0,5 + 2,9 + 9,5 + 3,1 + 33,1 + 5,7 + 4,7$ $\checkmark$MA $= 83,5$ OR/OF Total value / <i>Totale waarde</i> $\checkmark$RT $= 1\ 199,5 - 1\ 116$ $\checkmark$MA $= 83,5$	1RT correct values 1MA adding OR/OF 1RT both correct values 1MA subtracting (2)	D L1 E
3.1.2	The table value is given in ten thousands. $\checkmark\checkmark$O <i>Die tabelwaarde is gegee in tien duisende.</i> OR/OF Rounding issues / <i>Probleme met afronding</i> $\checkmark\checkmark$O	2O difference in table value from actual value OR/OF 2O rounding (2)	D L4 M
3.1.3	Number of people / <i>aantal mense</i> $\checkmark$RT $= 365,9 \times 10\ 000$ $= 3\ 659\ 000$ OR/OF 365,9 ten thousand / <i>tien duisend</i> $\checkmark$A	1RT reading from table 1A correct value (2)	D L1 E
3.1.4	Medical sector e.g. doctor/ nurse Security sector e.g. police / security guards Essential services e.g. cashier Construction sector e.g. plumbing / electrician / builder Agricultural sector e.g. farming <i>Mediese sektor bv. dokter / verpleegster</i> $\checkmark\checkmark$A <i>Sekuriteit sektor bv. polisie / sekuriteitswag</i> <i>Essensiële dienste bv. kasier</i> <i>Konstruksie sektor bv. loodgieter / elektrisiën / bouer</i> <i>Landbou sektor bv. boerdery</i>	2A correct job (2)	D L1 E

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
3.1.5	Mean in ten thousand / <i>Gemiddeld in tien duisend</i> ✓RT $= \frac{21,7 + 7,2 + 0,5 + 3,2 + 9,4 + 2,4 + 36,6 + 5,8 + 6,3}{9}$ 9 ✓MA $= \frac{93,1}{9}$ ✓S $= 10,34444$ Mean = 103 444,4 OR/OF 103 444 ✓CA $= \frac{1\,168,1 - 1\,075}{9}$	1RT adding correct values 1MA concept of mean 1S simplification 1CA correct mean NPR (4)	D L2 M
3.2.1	Quarter / <i>Kwartaal</i> 3 3 rd / <i>3de</i> ✓✓RT Third / <i>Derde</i>	2RT correct quarter (2)	D L1 E
3.2.2	Number of unemployed / <i>Aantal werkloos</i> ✓RT 7,6 million + 183 000 ✓C 7 600 000 + 183 000 = 7 783 000 ✓CA OR/OF ✓RT 7,6 million + 0,183 million ✓C = 7,783 million / <i>miljoen</i> ✓CA	1RT correct values 1C correct conversion 1CA simplification OR/OF 1RT correct values 1C correct conversion 1CA simplification (3)	D L2 M

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
3.2.3	✓RT ✓RT $Q1 = 34\%$; $Q3 = 37,4\%$ Increase % / Toename % $= 37,4\% - 34\%$ $= 3,4\%$ ✓CA	1RT quarter 1 1RT quarter 3 Accept quarter 3: $37,3\% - 37,5\%$ 1CA increase Accept: $3,3\% - 3,5\%$ (3)	D L2 M
3.2.4	Total number of people / <i>Totale aantal mense</i> $\frac{100}{34,4} \times 7,6$ million ✓RT $= 22\,093\,023,26$ $= 22\,093\,023$ ✓A ✓MCA Number of employed people $= 22\,093\,023 - 7\,600\,000$ $= 14\,493\,023$ ✓CA OR/OF ✓RT Employed $= 100\% - 34,4\% = 65,6\%$ ✓A ✓MCA Number of employed people $= \frac{65,6}{34,4} \times 7\,600\,000$ $= 14\,493\,023$ ✓CA	1RT correct percentage 1A simplification 1MCA subtracting values 1CA total number of people OR/OF 1RT correct percentage 1A simplification 1MCA ratio calculation 1CA total number of people NPR (4)	D L2 M
		[24]	

QUESTION/VRAAG 4 [33 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.1.1	Tax Payable (before rebates) $= R115\,762 + [36\% \times (\text{annual taxable income} - 488\,700)]$ <i>Belasting betaalbaar (voor korting)</i> $= R115\,762 + [36\% \times (\text{jaarlikse belasbare inkomste} - 488\,700)]$ ✓SF Tax payable $= R115\,762 + [\frac{36}{100} \times (495\,602 - 488\,700)]$ ✓MA $= R115\,762 + 2\,484,72$ ✓MA $= R118\,246,72$ ✓CA	AO 1SF substituting value 1MA adding values 1CA simplification NPR (3)	F L2 E
4.1.2	Monthly tax (before rebate) / <i>Maandelikse belasting (voor belastingkorting)</i> $= R118\,246,72 \div 12$ ✓MA $= R9\,853,89$ ✓A After rebate / <i>Na belastingkorting</i> ✓MA $= R9\,853,89 - R1\,368,75$ $= R8\,485,14$ ✓MCA Monthly taxable income (before rebate) / <i>Maandelikse belasting (voor belastingkorting)</i> $= R495\,602 \div 12$ $= R41\,300,17$ ✓A Tax payable (according to table) / <i>Belasting (volgens tabel)</i> $= R8\,491$ He is incorrect / <i>Hy is verkeerd</i> ✓O OR/OF	CA from Question 4.1.1 1MA dividing by 12 1A monthly tax 1MA subtracting rebate 1MCA finding tax after rebate 1A monthly income 1O conclusion OR/OF	F L4 D

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
4.1.2	Annual rebate / <i>Jaarlikse korting</i> $= R1\,368,75 \times 12 \checkmark \text{MA}$ $= R16\,425 \checkmark \text{A}$ Annual tax (after rebate) / <i>Jaarlikse belasting (na belastingkorting)</i> $= R118\,246,72 - R16\,425$ $= R101\,821,72 \checkmark \text{A}$ Monthly income / <i>Maandelikse inkomste</i> $= R495\,602 \div 12$ $= R41\,300,17 \checkmark \text{A}$ Annual tax / <i>Jaarlikse belasting</i> $= R8\,491 \times 12$ $= R101\,892 \checkmark \text{MCA}$ He is incorrect / <i>Hy is verkeerd.</i> $\checkmark \text{O}$ OR/OF Monthly tax before rebate / <i>Maandelikse belasting voor korting</i> $= R118\,246,72 \div 12 \checkmark \text{MA}$ $= R9\,853,89 \checkmark \text{A}$ Monthly taxable income / <i>Maandelikse belasbare inkomste</i> $= R495\,602 \div 12$ $= R41\,300,17 \checkmark \text{A}$ Before rebate / <i>Voor korting</i> $= R8\,491 + R1\,368,75 \checkmark \text{MA}$ $= R9\,859,75 \checkmark \text{MCA}$ He is incorrect / <i>Hy is verkeerd.</i> $\checkmark \text{O}$	1MA multiplying by 12 1A correct annual rebate 1A annual tax 1A monthly income 1MCA annual tax 1O conclusion OR/OF 1MA dividing by 12 1A correct answer 1A correct answer 1MA adding rebate 1MCA finding tax after rebate 1O conclusion	F L4 D (6)

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.1.3	Probability / <i>waarskynlikheid</i> 0 ; 0% ; impossible / <i>onmoontlik</i> / zero / <i>nul</i> ✓✓A	2A probability (2)	P L2 M
4.2.1	Two million five hundred and eighty four thousand one hundred and seventy six. <i>Twee miljoen vyf honderd vier en tagtig duisend een honderd ses en sewentig.</i> ✓✓A	2A correct words (2)	D L1 E
4.2.2	✓RT 407 739 : 61 934 : 36 085 ✓MA	1RT correct values 1MA correct order (2)	D L2 M
4.2.3	16 426; 18 235; 19 077; 21 887; 36 085 ✓A Median / <i>mediaan</i> = 19 077 ✓A	AO 1A arranging values 1A correct median (2)	D L2 E

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.2.4	Number of Ford F-Series / <i>Aantal Ford F-Reeks</i> $\checkmark\text{MA}$ $\checkmark\text{RT}$ $= 357\,243 - (53\,757 + 51\,684 + 73\,467 + 61\,934)$ $= 357\,243 - 240\,842$ $= 116\,401$ $\checkmark\text{CA}$	AO 1RT correct values from graph 1MA subtracting from total 1CA simplification (3)	D L2 E
4.2.5	Interquartile range / <i>Interkwartielomvang</i> $\checkmark\text{A}$ $\text{IQR} = Q3 - Q1$ $\checkmark\text{SF}$ $7\,669 = Q3 - 11\,408$ $\checkmark\text{MA}$ $Q3 = 7\,669 + 11\,408$ $= 19\,077$ $\checkmark\text{CA}$	AO 1A correct formula 1SF substituting into formula 1MA changing the subject of the formula 1CA simplification (4)	D L3 M
4.2.6	2020 Price of Ford F-Series / <i>2020 prys van Ford F-reeks</i> $\checkmark\text{RT}$ $\$32\,332 \times \frac{100}{107}$ $\checkmark\text{RT}$ $= \$30\,216,82$ $\checkmark\text{A}$ 2019 Price of Ford F-Series / <i>2019 prys van Ford F-reeks</i> $\$30\,216,82 \times \frac{100}{101,4}$ $\checkmark\text{MA}$ $= \$29\,799,63$ $\checkmark\text{CA}$ $\checkmark\text{O}$ The statement is not valid / <i>Die bewering is nie geldig nie.</i> OR/OF 2019 Price of Ford F-Series / <i>2019 prys van Ford F-reeks</i> $\checkmark\text{RT}$ $\checkmark\text{RT}$ $\$32\,332 \times \frac{100}{107} \times \frac{100}{101,4}$ $\checkmark\text{RT}$ $\checkmark\text{RT}$ $= \$29\,799,63$ $\checkmark\text{CA}$ The statement is not valid / <i>Die bewering is nie geldig nie.</i> $\checkmark\text{O}$	1RT numerator 1RT denominator 1A 2020 price 1MA concept of % decrease 1CA simplification 1O not valid OR/OF 1RT numerator 1RT denominator 1RT numerator 1RT denominator 1CA simplification 1O not valid (6)	F L4 D

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
4.2.7	Probability / Waarskynlikheid $\checkmark$ RT $\frac{569\,388}{2\,584\,176} \times 100\%$ $\checkmark$ RT $= 22,03\% \quad \checkmark$ CA	1RT correct numerator 1RT correct denominator 1CA simplification NPR (3)	P L2 M
		[33]	

QUESTION/VRAAG 5 [29 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.1.1	Western Cape / Wes-Kaap ✓✓RT	2RT correct province (2)	D2 L2
5.1.2	✓RT $3\% + 4\% + 4\%$ $= 11\%$ ✓A $= \frac{4}{11} \times R8\,300\,000\,000$ ✓MCA $= R3\,018\,181\,818 / R3,018181818 \text{ billion} / \text{miljard}$ ✓CA OR/OF Total contribution / Totale bydrae ✓RT $= \frac{100}{11} \times R8\,300\,000\,000$ $= R75\,454\,545\,454$ ✓A WC contribution / WK bydrae $= \frac{4}{100} \times R75\,454\,545\,454$ ✓MCA $= R3\,018\,181\,818$ ✓CA	1RT all 3 values 1A simplification 1MCA calculating ratio 1CA simplification OR/OF 1RT correct values 1A simplification 1MCA calculating ratio 1CA simplification NPR (4)	F L3
5.1.3	Transport / Vervoer ✓✓RT	2RT correct sector (2)	D2 L2
5.1.4	Finance / Finansies ✓✓RT	2RT correct sector (2)	D2 L3
5.1.5	Western Cape / Wes-Kaap ✓✓RT	2RT correct province (2)	D2 L2
5.2.1	Japanese yen / Japanese yen ✓✓RT	2RT correct currency (2)	F L1 M
5.2.2	Russian Rouble = $\frac{1}{6,97481}$ ✓A $= 0,143373$ ✓A	1A numerator 1A denominator (2)	F L1 E

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.2.3	$\begin{aligned} &\checkmark A \\ &= \frac{R1\ 230\ 000}{R1} \times CAD\ 0,084845 \checkmark MA \\ &= CAD\ 104\ 359,35 \checkmark A \end{aligned}$ OR/OF $\begin{aligned} &\checkmark A \\ &= \frac{R1\ 230\ 000}{R11,785} \times CAD\ 1 \\ &= CAD\ 104\ 369,9618 \checkmark MA \\ &= CAD\ 104\ 369,96 \checkmark A \end{aligned}$	1A correct exchange rate 1MA multiply with exchange rate 1A simplification OR/OF 1A correct exchange rate 1MA dividing by exchange rate 1A simplification NPR (3)	F L3 D
5.2.4	Diversify his income / <i>Diversifiseer sy inkomste.</i> $\checkmark\checkmark O$ OR/OF The Canadian currency is stronger / <i>Die Kanadese geldeenheid is sterker as die rand.</i> $\checkmark\checkmark O$ OR/OF He will get a better return on his investment / <i>Hy sal 'n beter opbrengs kry op sy belegging.</i> $\checkmark\checkmark O$	2O reason (2)	F L4 M

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.2.5	Amount interest / <i>Bedrag rente</i> Year / <i>jaar</i> 1 $R1\ 230\ 000 \times \frac{8,1}{100} \quad \checkmark \text{MA}$ $= R99\ 630 \quad \checkmark \text{A}$ Total after year 1 / <i>Totaal na jaar 1</i> $R1\ 230\ 000 + R99\ 630$ $= R1\ 329\ 630 \quad \checkmark \text{A}$ Year / <i>jaar</i> 2 $R1\ 329\ 630 \times \frac{8,1}{100}$ $= R107\ 700,03$ Total after year 2 / <i>Totaal na jaar 2</i> $R1\ 329\ 630 + R107\ 700,03$ $= R1\ 437\ 330,03 \quad \checkmark \text{CA}$ 8 months / <i>maande</i> $R1\ 437\ 330,03 \times \frac{8,1}{100} \times \frac{8}{12} \quad \checkmark \text{MA}$ $= R77\ 615,82162$ Final amount / <i>Finale bedrag</i> $= R1\ 437\ 330,03 + R77\ 615,82162$ $= R1\ 514\ 945,852 \quad \checkmark \text{CA}$ $R1\ 529\ 360 - R1\ 514\ 945,852$ $= R14\ 414,15 \quad \checkmark \text{CA}$ His statement is valid. / <i>Sy bewering is geldig.</i> ✓O	MA calculating 8,1% 1A interest year 1 1A amount end year 1 1CA amount year 2 1MA calculating 8 months 1CA final amount 1CA difference 1O conclusion	F L4 D

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
	OR/OF	OR/OF	
5.2.5	Total year 1 / <i>Totaal jaar 1</i> $\checkmark A$ $R1\ 230\ 000 \times 1,081 \checkmark MCA$ $= R1\ 329\ 630 \checkmark A$ Total year 2 / <i>Totaal jaar 2</i> $R1\ 329\ 639 \times 1,081$ $= R14\ 373\ 30,03 \checkmark CA$ Interest rate for 8 months / <i>Rentekoers vir 8 maande</i> $8,1\% \times 8 \div 12$ $= 5,4\% \checkmark A$ Total after 2 years 8 months / <i>Totaal na 2 jaar en 8 maande</i> $R1\ 437\ 330,03 \times 1,054$ $= R151\ 4\ 945,85 \checkmark CA$ Difference in interest earned / <i>Verskil in rente verdien</i> $R1\ 529\ 360 - R1\ 514\ 945,85$ $= R14\ 414,15 \checkmark CA$ The statement is valid / <i>Bewering is geldig.</i> $\checkmark O$	1A calculating 1,081 1MCA multiplying with 1,081 1A amount end year 1 1CA amount end year 2 1A calculating 5,4% 1CA final amount 1CA difference 1O conclusion NPR	
		(8)	
		[29]	
		TOTAL/TOTAAL: 150	